

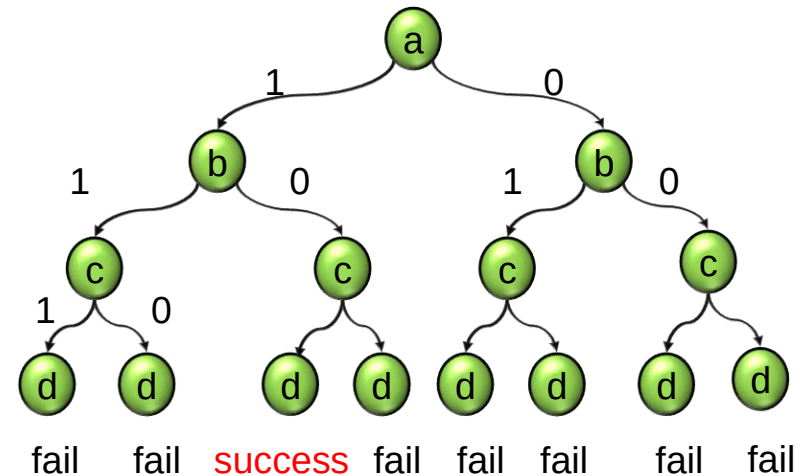
Combinational ATPG

- Introduction
- Deterministic Test Pattern Generation
 - ◆ Boolean difference *
 - ◆ Path sensitization **
 - ◆ D-Algorithm**
 - ◆ PODEM**
 - ◆ FAN**
 - ◆ SAT-based *
- Acceleration Techniques
- Concluding Remarks

Two ATPG categories:

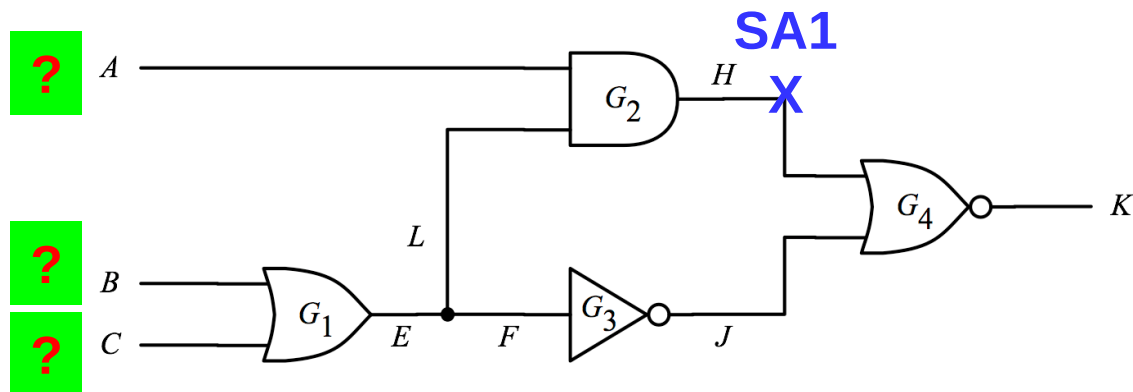
***Boolean-based methods**

****Path-based methods**



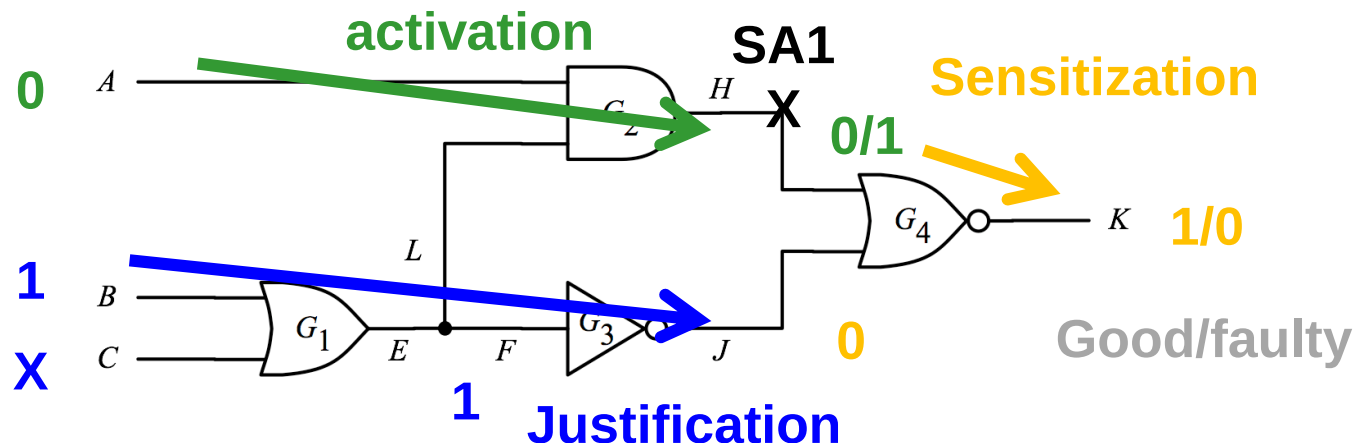
Motivating Problem

- We do not need to know Boolean expression
 - ♦ we can just find a test from circuit netlist



Let's Analyze What We Did

- **Fault activation:** Assign gate inputs to generate appropriate value at fault site (H)
 - ♦ $A=0$
- **Sensitization:** Assign side-inputs to non-controlling value to propagate fault effect forward
 - ♦ $J=0$
- **Justification:** Assign primary inputs to achieve desired values
 - ♦ $B=1$



Single Path Sensitization

- Single path sensitization (SPS) Algorithm:

- ① **Fault activation** (aka. **Fault excitation**)

Assign gate inputs to generate value at fault site

Desired value opposite to the faulty value (e.g. 0 for SA1)

- ② **Fault effect propagation**:

Select **one single path** from fault site to an output

Assign side inputs to sensitize fault effect along the path

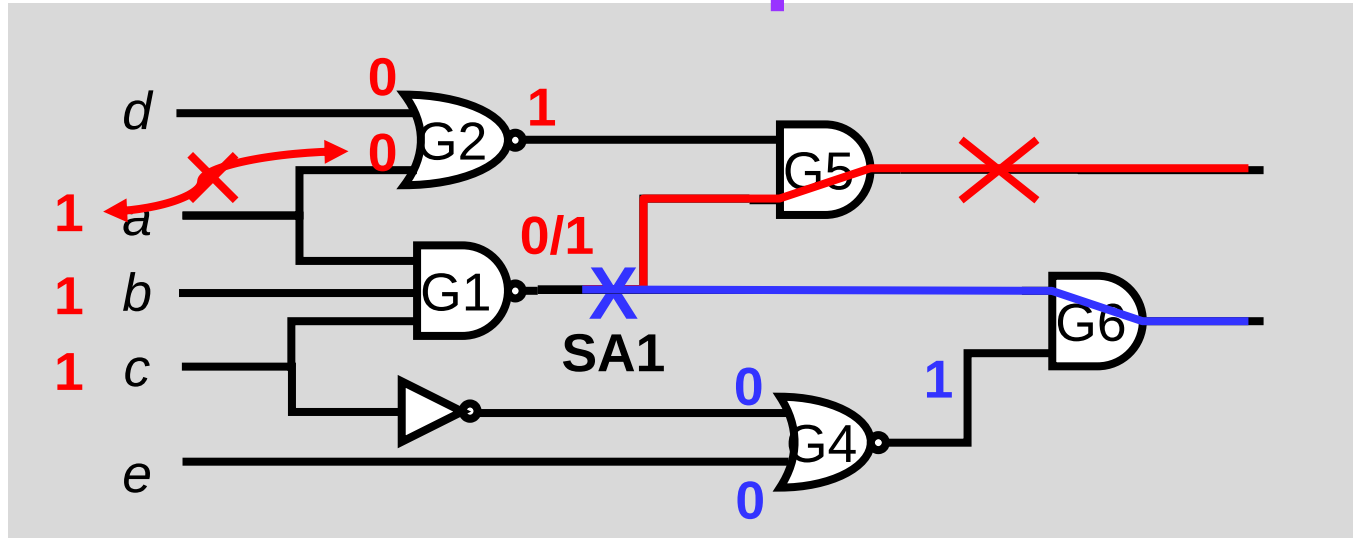
- ③ **Justification**:

Assign primary inputs to justify desired values assigned in

①&②

If justification fails, **backtrack**.

Example



Consider stuck-at-1 fault

① Fault activation

♦ $a = b = c = 1$

② Fault effect propagation: two propagation paths

♦ Choose path **{G5}**. Want $G2 = 1$

③ $a = d = 0 \rightarrow$ **justification fails!**

② Backtrack! Choose another path **{G6}**. Want $G4 = 1$

③ $c = 1, e = 0 \rightarrow$ **justification succeeds**

♦ test pattern: $abce'$ generated

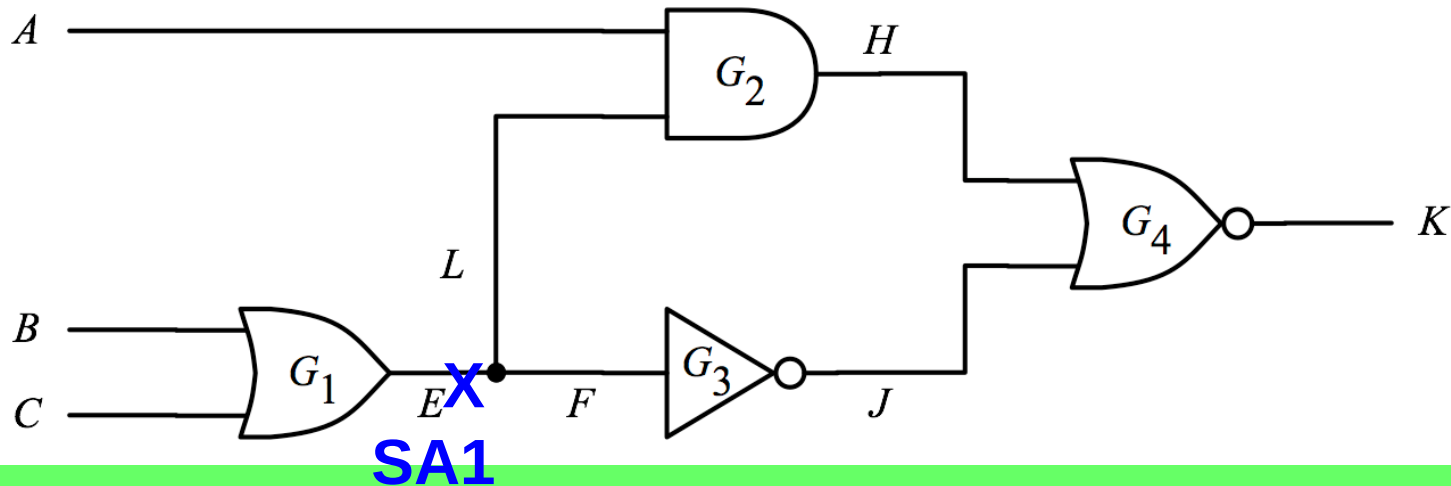
Quiz

Q1: Generate a test pattern for *E* SA1 fault. Choose path *ELHK*.

A:

Q2: (Cont'd) Backtrack to another path *EFJK*.

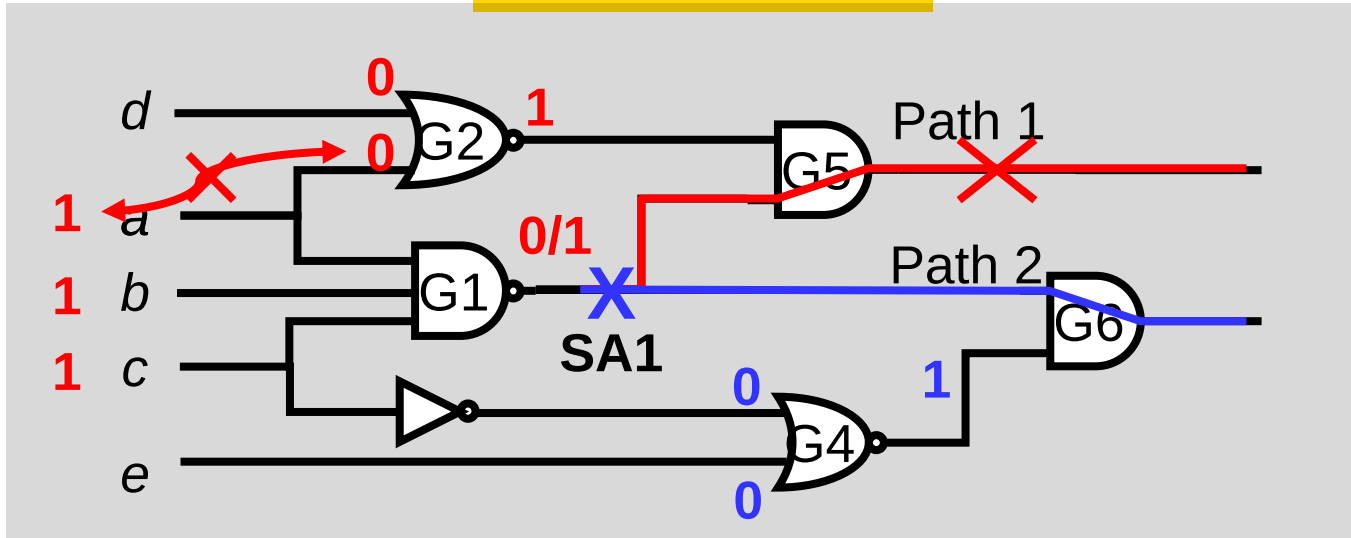
A:



Important to Choose Correct Path

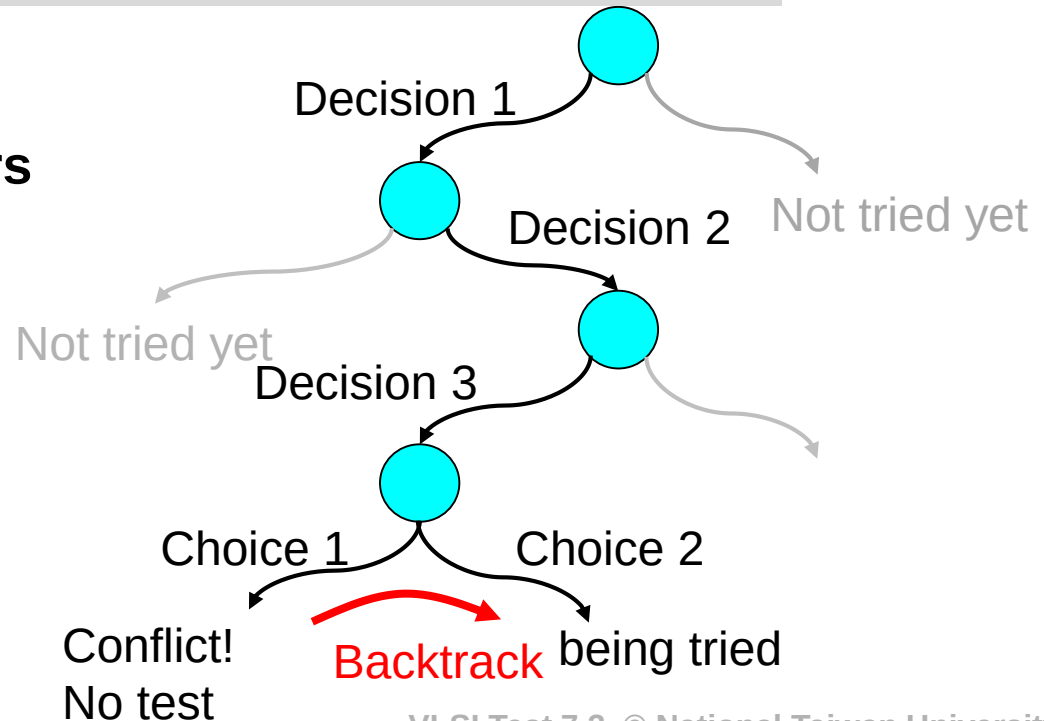
But it is difficult...

Backtrack



- When we made a mistake in decision tree, **conflict** occurs

- Go back to a previous decision point
- Change decision
- Redo the rest



Pros and Cons

😊 Pros

- ◆ Easy to implement
- ◆ No Boolean equation needed

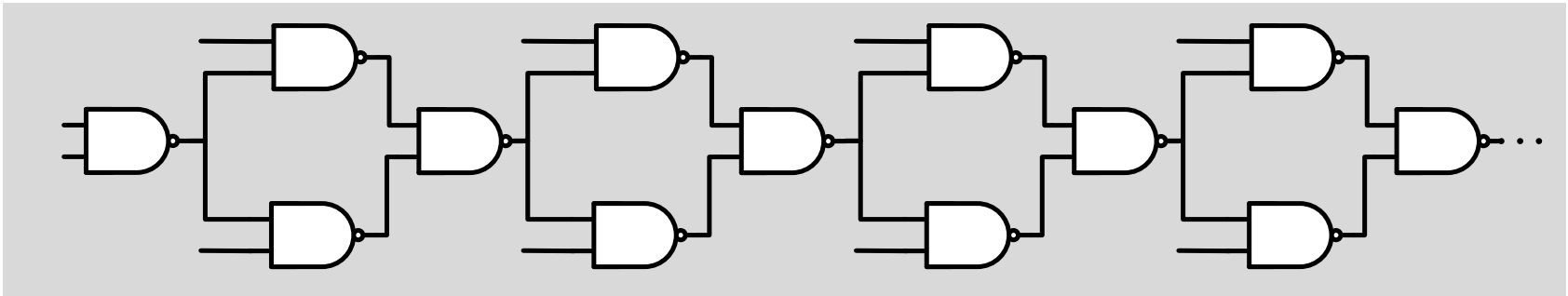
😞 Cons

- ◆ Q1: Too many paths to choose, which one is correct?
- ◆ Q2: Single-path sensitization not enough to detect all faults

- Single path sensitization (SPS) Algorithm:
 - ① **Fault activation** (aka. **Fault excitation**)
Assign internal signals to generate value at fault site
Desired value opposite to the faulty value
 - ② **Fault effect propagation**:
Select **one single path** from fault site to an output
Assign internal signals to sensitize fault effect along the path
 - ③ **Justification**:
Assign primary inputs to justify signal values assigned in ①&②
If justification fails, **backtrack**.

Q1: Too Many Paths !

- How to choose correct path?
 - ♦ No smart algorithm
 - ♦ Simple idea: exhaustively try all paths
- How many paths in a circuit?
 - ♦ Worst case example: $3n$ gates, 2^n paths! ($n = \#$ of stages)

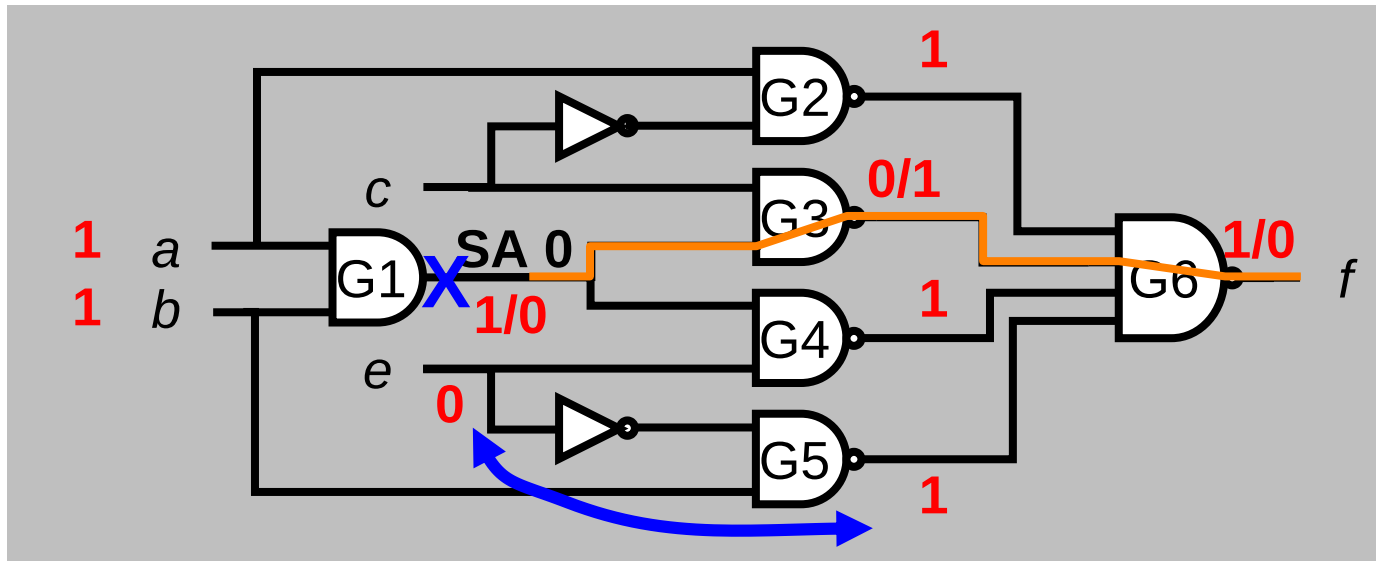


of Paths is Exponential to Circuit Size!
Impossible to Try All

Q2: Single Path Not Enough

- Example

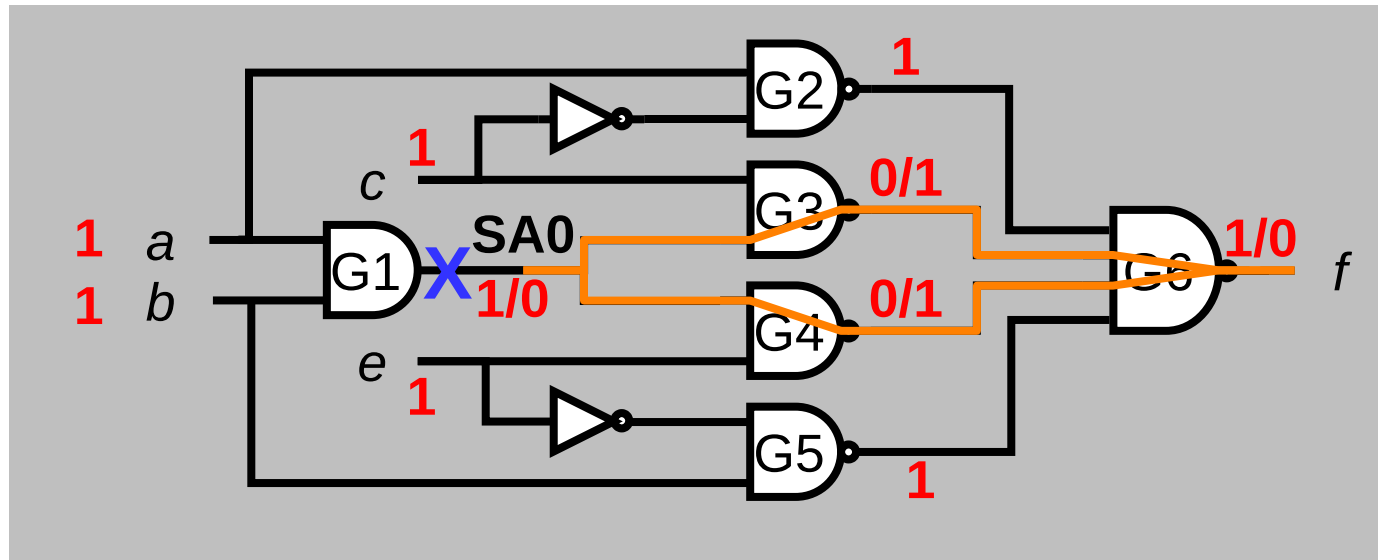
- ① Fault activation: $a = b = 1$
 - ② Fault propagation: Choose path **{G3-G6}**. want $G2 = G4 = G5 = 1$
 - ③ $G4 = 1 \rightarrow e = 0 \rightarrow G5 = 0 \rightarrow$ justification fails
 - ② Choose another path **{G4-G6}**. Justification also fails.
- ◆ SPS algorithm fails



This Fault Is Actually Testable. Why SPS Fail?

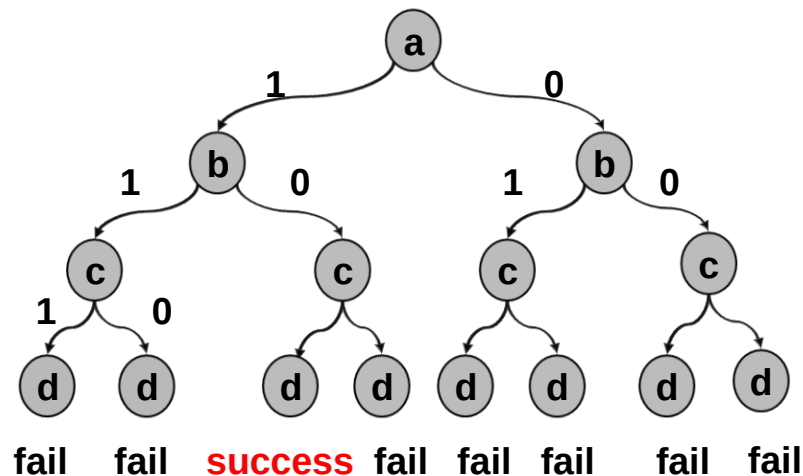
Solution: Multiple Path Sensitization

- This fault requires *multiple path sensitization*
- Both two paths {G3-G6} and {G4-G6} are sensitized
 - ♦ Error is propagated along both paths simultaneously
 - ♦ $G2 = G5 = 1$, $c = e = 1$
 - ♦ Test generated successfully
- FFT: can we extend 1-path to 2, 3, 4, 5...path sensitization?



Complete ATPG Algorithm

- A **complete** ATPG exhausts the whole input space (2^n)
 - ♦ If a test pattern exists, complete ATPG will find it **for sure**
- An **incomplete** ATPG does **NOT** exhaust the whole input space
 - ♦ ATPG may fail even though a test pattern **DOES** exist



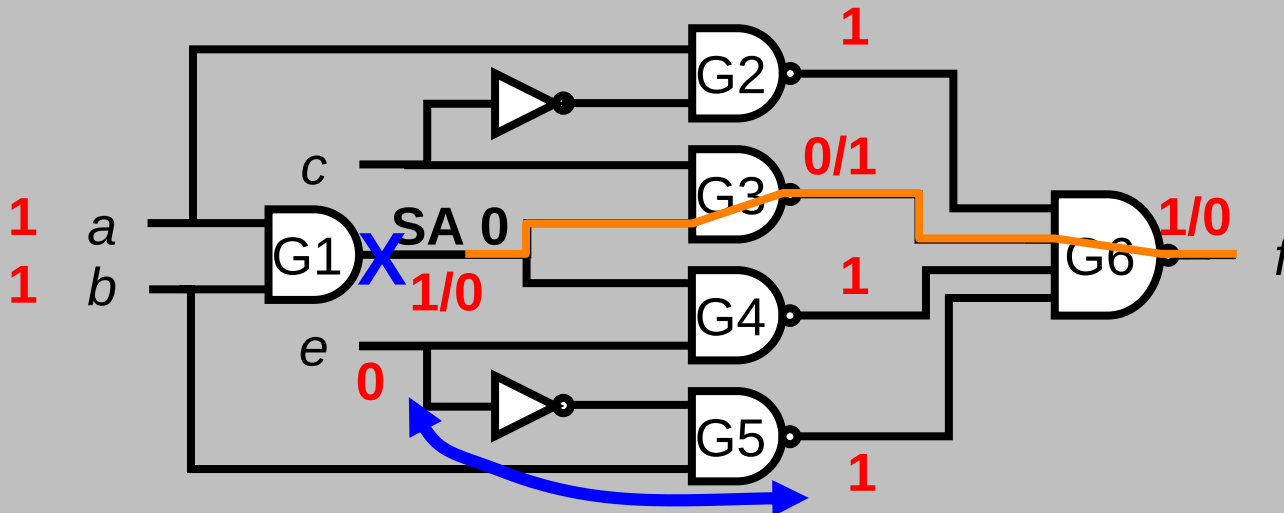
**Complete ATPG Guarantees to Find Solution
if it exists**

Quiz

Q: Is Single Path Sensitization a **complete ATPG algorithm**?

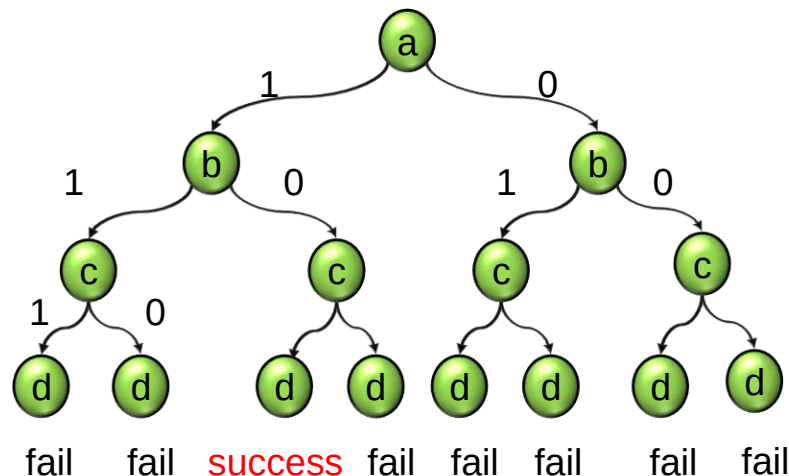
Hint: recall this example

- Single path sensitization (SPS) Algorithm:
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If justification fails, **backtrack**.



Summary

- Single Path Sensitization Algorithm
 - ① Fault activation
 - ② Fault effect propagation
 - ③ Justification
- Complete ATPG: guarantee to find solution if one exists
- Disadvantages of SPS algorithm
 - ◆ Number of paths is **exponential** to circuit size. Too many!
 - ◆ Single path sensitization is **incomplete** ATPG



FFT

- Q1: If SPS fails, can we extend 1-path to 2, 3, 4, 5... paths?
 - ♦ What is the problem for multiple path sensitization algorithm?
- Q2: Pros and Cons of complete ATPG algorithm?
 - ♦ in terms of run time and fault coverage

